

Ayurvedic Health and Wellness Navigator Using Retrieval Augmented Generation

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Abstract— Ayurveda, an ancient system of medicine, is experiencing rapid global adoption, driving the need for accessible and reliable health guidance. However, traditional Ayurvedic texts, often written in Sanskrit, are complex and require advanced interpretation for accurate application. This project aims to address these challenges by integrating Generative AI with Retrieval-Augmented Generation, leveraging authentic Ayurvedic texts to provide more precise and contextually relevant guidance. By evaluating various large language models and embedding models, such as OpenAI and MiniLM, the project aims to enhance the effectiveness of retrieving and generating Ayurvedic knowledge. A key innovation of the project is the development of a speech-to-speech AI bot, utilizing Deepgram's speech-to-text and text-to-speech capabilities to offer users a seamless, personalized, and interactive experience. It was observed that the combination of OpenAI Embeddings with GPT-3.5 Turbo, achieving Answer relevancy and Context recall scores of 0.94 and 0.73 respectively, delivered the best performance in developing the speech-to-speech bot. This bot serves as a comprehensive wellness navigator, providing Ayurvedic guidance tailored to individual health needs. The future plans include integrating authentic Sanskrit Ayurvedic texts to further refine the system's accuracy and personalization.

Keywords— Natural Language Processing; Word Embedding; Retrieval Augmented Generation; Chatbot; OpenAI; Gemini.

I. INTRODUCTION

Ayurveda, an ancient holistic system of medicine originating from India, has been practiced for over 3,000 years and is gaining increasing global attention. Its comprehensive approach focuses on maintaining balance within the body, mind, and spirit to promote health and well-being. As more people around the world seek alternative and natural healing methods, the demand for accessible, reliable Ayurvedic guidance is rising. However, one of the major challenges lies in interpreting and applying the vast knowledge embedded in traditional Ayurvedic texts. These texts, often written in Sanskrit and other ancient languages, contain rich metaphors, intricate concepts, and detailed medical knowledge that require expert understanding for accurate use. To make this wealth of knowledge accessible to modern users, there is a pressing need to integrate these ancient teachings with advanced technological systems that can break down the barriers of language and context.

This research integrates generative AI with Retrieval-Augmented Generation (RAG) [1][2] and authentic Ayurvedic texts to develop an intelligent chatbot capable of providing precise, personalized, and contextually relevant Ayurvedic advice. The RAG pipeline, backed by a vector

database, enables dynamic retrieval of Ayurvedic knowledge, ensuring accuracy beyond static, script-based approaches. This adaptability is crucial for Ayurveda, where recommendations are tailored to an individual's Prakriti (constitution) and health condition. The system also incorporates advanced AI models and document search algorithms [3], along with a user-friendly speech-to-speech interface. Additionally, it evaluates multiple LLMs, including OpenAI, MiniLM, and BGE, to determine the optimal model for embedding and generating Ayurvedic knowledge, enhancing system reliability and performance.

In addition to the text-based model, this research introduces an innovative speech-to-speech AI bot designed to offer a seamless and personalized user experience. By integrating Deepgram's speech-to-text and text-to-speech technologies [4], the system enables natural voice interaction, allowing users to ask questions and receive responses in a conversational manner. Speech-to-speech AI in healthcare enables natural, real-time patient interactions, with recent advancements enhancing personalization, affective computing, and generative AI for better engagement [5]. This feature also supports multiple languages and accents, making it adaptable for a global audience. This work seeks to bridge the gap between ancient Ayurvedic wisdom and modern technology by developing an AI-powered, speech-to-speech chatbot that leverages RAG and ancient Ayurvedic wisdom [6]. Integrating advanced language models, a vector-based retrieval system, and Deepgram's speech technologies, the system interprets complex Sanskrit-based knowledge and delivers personalized advice in a natural, conversational format. Each recommendation, whether related to diet, exercise, herbal remedies, or stress management, is tailored to the user's unique health profile. By combining ancient traditions with cutting-edge AI, this project empowers users with reliable, accessible, and holistic health solutions. Use the enter key to start a new paragraph. The appropriate spacing and indent are automatically applied.

The key contribution of this work is given as follows:

- Comparative Analysis of Embedding Models and Large Language Models (LLMs), evaluating BGE Embeddings, MiniLM embeddings, and other architectures for enhanced response accuracy.
- Implements RAG with Facebook AI Similarity Search (FAISS) [7] for optimized semantic search and precise retrieval of Ayurvedic knowledge from authentic texts.

- Conduct empirical evaluations using RAGAS, ensuring response quality and relevance through systematic performance benchmarking.
- Speech-to-speech Conversational AI enabling real-time, natural interactions for Ayurvedic consultations using Deepgram for STT and TTS.
- Develop an intuitive, real-time interface using Streamlit [8] facilitating seamless voice-based interactions and document-based responses.

The work explores various components of research work. Section 2 provides a review of existing research in the field of sentiment analysis followed by Section 3 that describes the dataset utilized in this study including details about its source, size, and features. Section 4 presents the proposed model for this research, explaining its architecture and components. Section 5 showcases the study's results, including statistical analyses. Finally, Section 6 summarizes the findings and offers recommendations for future research directions.

II. LITERATURE SURVEY

This section details the existing research work that has been carried out till now that focuses on the methodologies applied in the field of NLP in medical domain. The key points explored here include the use of embeddings for various tasks, different types of LLMs implemented, the choice of vector store, technologies employed and different kind of evaluation metrics. The review aims to highlight the research gaps in the existing studies.

The research conducted by S. K. et al. [9] explores an AI-driven chatbot system using NLP, RAG, and TTS to provide accurate guidance on infectious diseases, particularly COVID-19. Here the user queries effectively with the ensemble retriever-based RAG excelling in accuracy and the auto-merging retriever-based RAG performing better in contextual understanding than the standard naïve approach. Meanwhile, Su and Yang [10] examined the integration of RAG with the Llama model to reduce bias against traditional Chinese medicine (TCM) that enhanced response accuracy, nuance and reliability. Another approach by Yagnik et al.

[11] examined general and medical-specific distilled LMs for medical question-answering (QA). It compared fine-tuned domain-specific LMs and different architectures, finding decoder-only models like GPT superior. Additionally, dynamic prompting outperformed static methods. This study also highlighted the limitations of quantitative metrics like BLEU and ROUGE, as human evaluations often differ. Hamman et al. [12] introduced SelfRewardRAG, a framework integrating LLMs with RAG to mitigate knowledge obsolescence in healthcare. In this work it dynamically retrieved the latest information from sources like PubMed and incorporated iterative self-evaluation. SelfRewardRAG enhanced the accuracy and timeliness of medical QA. Experimental results showed that it outperformed the state-of-the-art models, demonstrating its potential to improve AI-driven healthcare applications.

REMED [13], a medical document retrieval framework incorporating dataset construction, embedding fine-tuning, RAG, and human evaluation. A key innovation, EM-FT, optimizes medical sentence representations in large pre-trained models, enhancing retrieval accuracy. The work carried out by Setty et al. [14] tackled financial document

retrieval challenges using advanced chunking, query expansion, metadata annotations, re-ranking, and domain-specific embedding fine-tuning and improving retrieval relevance. The model proposed by Fathullah et al. [15] which is a multilingual speech recognition model where LLMs were conditioned on audio embeddings from a pretrained encoder. This method outperformed monolingual systems and significantly boosted ASR robustness. RAPTOR [16] is a hierarchical tree-based retrieval system for efficiently handling lengthy documents. It enabled accurate extraction of contextually relevant information by leveraging structured information flow.

Šarčević et al. designed a generative AI platform [17] for PDF summarization that used OpenAI language models and Chroma vectors within a Gradio-powered interface. The system enhanced summarization through conversational retrieval chains and user-friendly interaction. Additionally, the platform employed conversational retrieval chains to provide contextual relevance, contributing to the overall success of the summarization process. While the research carried out by Mohammed et al. [18] demonstrated that RAG could enhance language models better than fine-tuning. The Mistral-7B-OpenOrca model achieved a high mean human rating and efficient response times, suggesting practical deployment potential for real-world use. Another work [19] introduced Joint Medical LLM and Retrieval Training (JMLR), a novel approach that jointly trains an LLM and a retrieval system for medical QA. The JMLR-13B model outperformed a 70B model, demonstrating superior reasoning and fewer hallucinations, even surpassing Claude3-Opus. Zhao et al. [20] reviewed RAG for multimodal inputs like images, code, structured knowledge, audio, and video. It highlights multimodal RAG's role in advancing generative AI and calls for further exploration. Hybrid generation models explored by Lewis et al. [21] combined parametric and non-parametric memory, achieving state-of-the-art open-domain QA results. User evaluations showed that RAG-based systems generate more factual responses than parametric models like BART. Finally, Sanjay S et al. introduced an Ayurvedic chatbot [22] that merges traditional medical knowledge with modern AI to deliver personalized recommendations. Their work detailed key implementation challenges and future enhancement opportunities. Despite significant advancements in the integration of LLMs and RAG for specialized domains like healthcare and finance, limited research has focused on speech-to-speech systems that combine traditional knowledge systems such as Ayurveda with modern AI techniques.

While existing works emphasize multimodal RAG, hybrid memory models, and domain-specific fine-tuning for textual tasks, the gap remains in adapting these methodologies for real-time, speech-driven interfaces in traditional medicine. Furthermore, the comparative analysis of embeddings and LLMs for optimizing performance in speech-based applications is not explored. To address these gaps, this research work focuses on developing a speech-to-speech AI bot for Ayurveda that integrates Deep gram for speech processing and conducting a comprehensive performance evaluation of various embedding models and LLMs.

III. DATA ACQUISITION AND PREPARATION

The data for this work consists of textbooks in the PDF format, which have been obtained for analysis and

implementation purposes. The books are downloaded from [23], which consists of Ayurvedic materials ensuring accessibility for further development and research. Table I discusses about the books and their statistics used in this work. These books encompass a wide array of details about Ayurveda including foundational principles, practices, and remedies. The rich and diverse content from these texts serves as a comprehensive resource. By leveraging these books, the system can provide accurate insights and answers, making it an invaluable tool for tasks like question answering, summarization, and context-aware information retrieval.

TABLE I. BOOKS USED FOR KNOWLEDGE BASE AND WORD COUNT

Book Name	Total pages	Word count
Ayurveda for dummies	291	78828
Ayurveda The Science of Life (Common complicated diseases).	148	46943
Rasayana: Ayurvedic Herbs for Longevity and Rejuvenation	369	118229
The Hidden Secret of Ayurveda	71	18485
Scientific Basis for Ayurvedic Therapies	600	262016
Secrets of Indian Medicine	254	64584
Practice of Ayurveda	283	69896
The Ayurveda Encyclopaedia	688	312274

The data for the sample question and answer dataset is sourced from two renowned books: "Ayurveda in the Age of Biomedicine: Discursive Asymmetries and Counter-Strategies" [24] and "The Complete Book of Ayurvedic Home Remedies," [25] both in PDF format. A set of 50 questions covering different Ayurveda topics was compiled to create the ground truth data for RAG evaluations. Table II contains examples of questions created with the ground truth.

TABLE II. SAMPLE QUESTIONS WITH ANSWERS

Question	Ground Truth
What is the main cause for hypertension and remedies for it.	First, excess mental stress (i.e., worry, anxiety, tension) can be caused by Vayu. Second, certain hormonal imbalances (e.g., adrenal, thyroid, and reproductive hormones) can be caused by Pitta....
What are the home remedies for Angina (chest pain)?	Angina, or to give it its full name, angina pectoris (chest pain), is a condition created by kapha dosha. Accumulated kapha blocks the flow of prana vata into the coronary artery, so that the heart muscles do not receive sufficient blood and oxygen supply. It is a kind of local anemia....

IV. METHODOLOGY

The Ayurvedic Chatbot is designed to integrate speech-to-speech with the help of generative AI and various LLMs and embedding techniques. This chatbot has several significant modules as represented in the Fig 1. Each of these components are detailed below:

A. User interface and uploading PDF file

The user interface is designed to allow users to upload their data files through the Ayurvedic chatbot. Users can upload PDFs containing Ayurvedic texts, which are then processed

by the system for content extraction. This interface enables users to interact with the chatbot through both text and voice.

B. Document Ingestion Pipeline

This module processes and converts the uploaded PDF text into a machine-readable format through two main tasks. The first task is to extract the text with the help of PyPDF2 to retrieve raw text from the uploaded PDFs, which accounts for structural elements like headers, footers, and images. Second includes the division of text into smaller, manageable chunks to optimize processing and handle large documents effectively. To preserve contextual continuity between chunks, an overlap is introduced during the chunking process, which is set to 1000 by default as a hyperparameter. This can be adjusted based on the specific context. This step is crucial for ensuring seamless integration in subsequent semantic search and embedding tasks.

C. Text Embedding

Followed by the chunking process, the next stage is converting the processed text into numerical representations using embedding models. For this study, the various embedding techniques employed are OpenAI Embeddings, MiniLM, and BGE to generate dense vector representations that capture semantic meaning. This enables efficient semantic search beyond basic keyword matching. The size of each embedding model is detailed in Table III. Each text chunk is independently embedded, allowing scalable processing of large documents.

TABLE III. SUMMARY OF EMBEDDING TECHNIQUES AND PARAMETERS

Embeddings	Embedding Dimension
OpenAI	1536
BGE Embeddings	384
Mini LM Embeddings	384

D. Vector Store

The embeddings generated from the three models—OpenAI Embeddings, MiniLM, and BGE—are stored in separate local vector databases. For this purpose, FAISS has been selected due to its robust performance in handling large-scale similarity searches. Each vector store is indexed using cosine similarity and structured to allow fast and accurate retrieval of relevant information. By maintaining dedicated local stores for all three embedding models, the system ensures efficient handling of extensive datasets and delivers contextually relevant answers. This architecture enhances both scalability and response time, ensuring the Ayurvedic chatbot delivers real-time, accurate results.

E. Query Embedding and Semantic search

Once the vector stores are set up, the system becomes capable of handling user queries efficiently. When a user submits a query, it is embedded using the same model used during document embedding to maintain semantic consistency. The resulting query embedding is then compared against stored document vectors using semantic search techniques. This enables the retrieval of the most contextually relevant information, based on the meaning of the query rather than mere keyword matching, thereby enhancing the quality of the results.

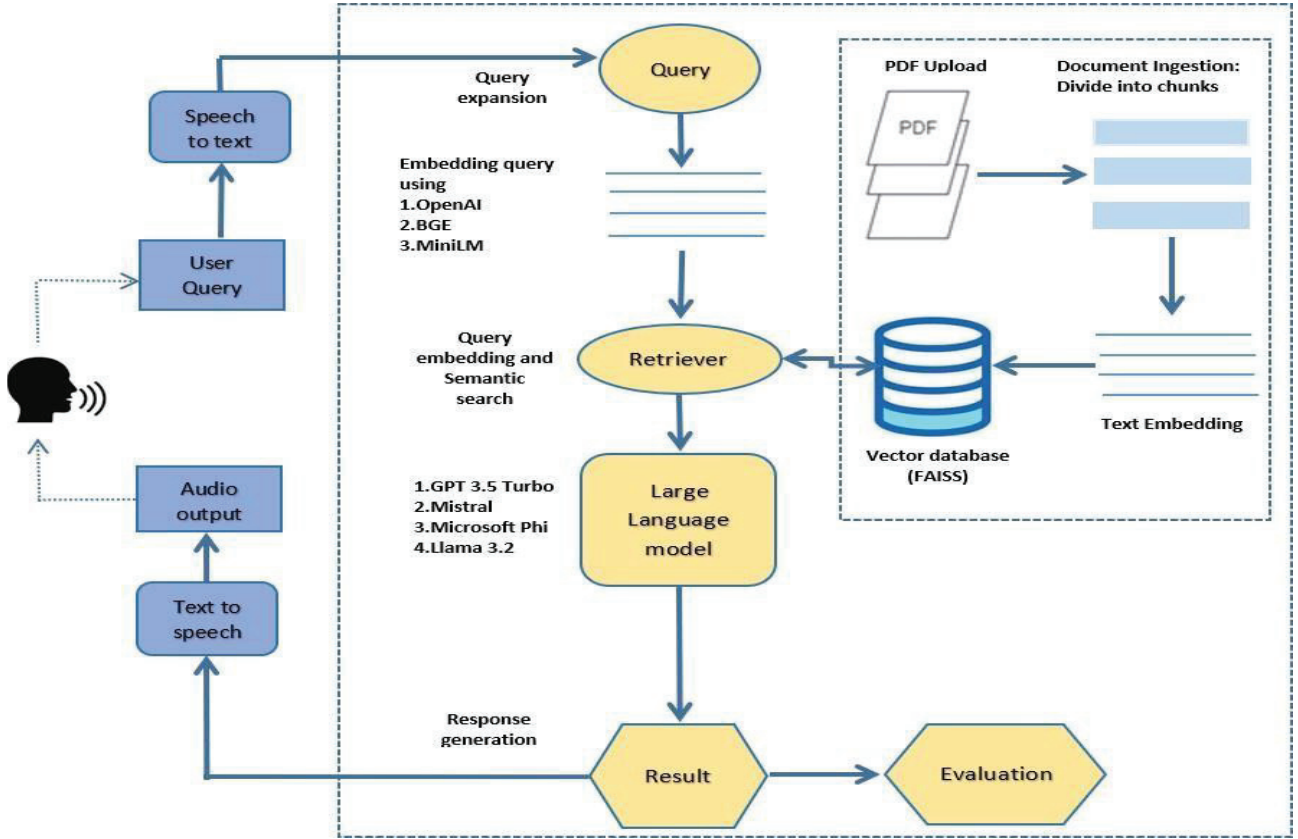


Fig. 1. Proposed Framework for Ayurvedic Chatbot using RAG

F. Response Generation

After retrieving the relevant contextual information from vector stores, the system proceeds to generate a response through a RAG pipeline. The extracted context is passed to multiple LMs including GPT-3.5 Turbo, Mistral, Microsoft Phi-3.5 and LLaMA 3.2-1B for response generation and comparative analysis. This multi-model approach ensures not only accuracy but also contextual alignment with Ayurvedic principles, while offering insights into the performance of various LLMs under the same information set.

G. Speech to Speech AI bot

To further enhance user interaction, the system incorporates a speech-to-speech AI bot that enables fully voice-based communication. This module leverages Deepgram's API for both speech-to-text (STT) and text-to-speech (TTS) processing. Real-time audio input is captured using Deepgram's Live Transcription API, which accurately converts spoken queries into text. Once the response is generated by the selected LLM, it is transformed back into natural-sounding speech using Deepgram's Aura Athena model. The system also includes a live microphone stream to support uninterrupted dialogue, allowing users to interact continuously without manual input. This voice-based interface delivers a highly intuitive and immersive experience, enabling users to receive Ayurvedic guidance through both voice input and

voice output. By supporting natural conversation, the chatbot becomes significantly more accessible and user-friendly, especially for users who prefer or rely on audio-based interaction.

H. Hyper parameter and evaluation metrics.

The key hyperparameters used in the development of Ayurvedic chatbot is outlined in Table IV. This table consists of parameters like chunk size, overlap, text splitting method and model settings for retrieval and processing. These parameters ensure efficient text segmentation, retrieval accuracy and optimal speech processing for the chatbot.

The evaluation metrics used in this work are discussed below:

Context Precision: It measures the proportion of retrieved context that is actually relevant to answering the query. A higher precision in (1) indicates that most of the retrieved information is useful and directly contributes to the response. The score ranges from 0 to 1.

$$\text{Context precision@K} = \frac{\sum_{k=1}^K (\text{Precision@k} \times v_k)}{\text{Total number of relevant items in the top K results}}$$

$$\text{Precision@k} = \frac{\text{true positives@k}}{(\text{true positives@k} + \text{false positives@k})} \quad (1)$$

where K is the total number of chunks in retrieved contexts and $v_k \in \{0,1\}$ is the relevance indicator at rank k.

Context Recall: This metric (2) evaluates the extent to which the necessary information for answering a query has been successfully retrieved. A high recall score, ranging from 0 to 1, indicates that the system is retrieving enough relevant context to produce a complete and well-informed response.

$$\text{Context recall} = \frac{\text{Number of claims in the reference supported by the retrieved context}}{\text{Total number of claims in the reference}} \quad (2)$$

TABLE IV. HYPERPARAMETER DETAILS FOR THE RAG BASED

AYURVEDIC CHATBOT	
Parameters	Value
Chunk size	1000 with an overlapping of 100
Chain type	stuff
Splitter	recursive text splitter
temperature	.7
model	nova-2
punctuate	True
language	en-US
encoding	linear16
channels	1
sample_rate	16000

Faithfulness: This measure is used to assess whether the generated response remains true to the retrieved context without introducing incorrect or misleading information. A high faithfulness score (3) means that the response is well-grounded in the provided evidence. The score ranges from 0 to 1.

$$\text{Faithfulness score} = \frac{\text{Number of claims in the response supported by the retrieved context}}{\text{Total number of claims in the response}} \quad (3)$$

Answer Relevancy: This metric (4) determines how well the generated response aligns with the user's query in terms of meaning and usefulness. A relevant answer, with a score ranging from 0 to 1, directly addresses the question and offers meaningful insights.

$$\text{Answer relevance} = \frac{1}{N} \sum_{i=1}^N \text{cosine similarity}(E_{g_i}, E_o)$$

$$\text{Answer relevancy} = \frac{1}{N} \sum_{i=1}^N \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\| \|E_o\|} \quad (4)$$

where:

E_{g_i} : Embedding of the i th generated question.

E_o : Embedding of the user input.

N: Number of generated questions.

V. RESULT AND DISCUSSION

The Ayurvedic chatbot has been developed along with the voice integration. This study aims to assess its performance for different LLMs and embedding models that have been utilized for building the Ayurvedic chatbot. The different LLMs, such as GPT-3.5, Microsoft Phi, LLaMA and Mistral were checked on their accuracy and/or contextual response generation proficiency. The performance of each model was further assessed with reference to measures of accuracy of response along with their applicability of the model to Ayurvedic ideas and its usability. These evaluations were performed using a particular method called RAGAS, which is used for evaluating RAG-based systems.

Table V provides an analysis of different embedding models and LLM combinations based on context precision, recall, faithfulness, and answer relevancy. OpenAI embeddings consistently achieve the highest precision and recall across various LLMs. When paired with GPT-3.5 Turbo, they yield a context precision of 0.935 and a recall of 0.7304, along with strong faithfulness of 0.921 and answer relevancy of 0.9444, making this combination the best performing.

Among all the LLMs used to evaluate, GPT-3.5 Turbo proves the most reliable, consistently maintaining high scores across all the embeddings. Mistral AI, when used with OpenAI embeddings, achieves a higher recall of 0.7837 than GPT-3.5 Turbo but has a lower faithfulness score of 0.5986, indicating occasional inaccuracies. Microsoft Phi 3.5 offers a balanced trade-off, achieving a recall of 0.7848 and a faithfulness score of 0.852, making it a strong alternative for generating accurate responses.

The performance of LLaMA 3.2 1B varies across embedding models while it achieves high answer relevancy at 0.9474 with OpenAI embeddings and its low faithfulness score of 0.2174 indicates reliability issues. BGE embedding perform well particularly with GPT-3.5 Turbo reaching strong faithfulness at 0.9077 and answer relevancy at 0.9483, though they fall slightly behind OpenAI embeddings in precision and recall.

TABLE V. COMPARISON OF PERFORMANCE ACROSS DIFFERENT LLM AND EMBEDDING TECHNIQUES

Embedding	LLM	Context precision	Context recall	Faithfulness	Answer relevancy
OpenAI	GPT 3.5 Turbo	0.935	0.7304	0.921	0.9444
	Mistral AI	0.9233	0.7837	0.5986	0.7367
	Microsoft Phi 3.5	0.9133	0.7848	0.852	0.8518
	Llama 3.2 1B	0.9161	0.787	0.2174	0.9474
BGE Embedding	GPT 3.5 Turbo	0.8794	0.6893	0.9077	0.9483
	Mistral AI	0.8983	0.6826	0.6312	0.6656
	Microsoft Phi 3.5	0.8922	0.6828	0.8602	0.8807
	Llama 3.2 1B	0.89	0.7059	0.6607	0.8314
MiniLM embedding	GPT 3.5 Turbo	0.8907	0.5996	0.9125	0.9351
	Mistral AI	0.8039	0.5621	0.6003	0.5494
	Microsoft Phi 3.5	0.8022	0.5265	0.7306	0.922
	Llama 3.2 1B	0.7854	0.5689	0.4692	0.8355

MiniLM embedding exhibits the weakest performance overall, especially with LLaMA and Mistral AI, where recall and faithfulness decline significantly. However, when paired with GPT-3.5 Turbo, MiniLM achieves reasonable faithfulness at 0.9125 and answer relevancy at 0.9351, suggesting acceptable performance with a strong LLM.

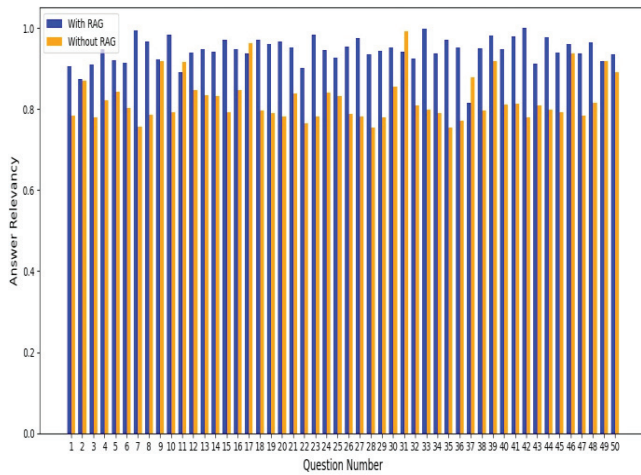


Fig. 2. Performance comparison of GPT 3.5 turbo with and without RAG with Open AI Embedding

Fig. 2 shows that **GPT-3.5 Turbo with RAG consistently outperforms the model without RAG in terms of answer relevancy across 50 questions.** The blue bars indicate performance with RAG shows higher and more stable relevancy scores that frequently reach the maximum of 1.0. This indicates an improved accuracy and contextual understanding when RAG is employed. In contrast, the orange bars, which indicate performance without RAG, exhibit more variability and lower scores, with significant gaps in performance for certain questions such as 7, 10, and 31, which highlight the limitations of the standalone model in retrieving contextually precise information. This confirms the effectiveness of RAG in enhancing the quality and reliability of responses.

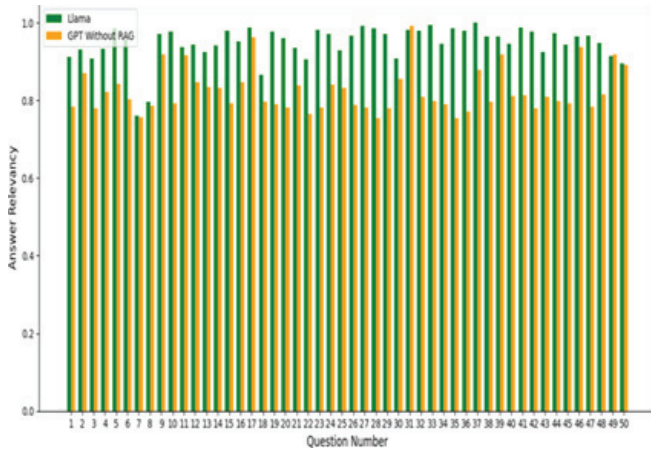


Fig. 3. Performance comparison of Llama 3.2 GPT 3.5 turbo without RAG with Open AI Embedding

Fig. 3 highlights that **Llama 3.2 with RAG, shown in green bars, outperforms GPT-3.5 Turbo without RAG, shown in orange bars, in terms of answer relevancy across 50 questions.** Llama 3.2 with RAG consistently achieves higher scores often nearing the maximum of 1.0, reflecting its ability to retrieve and utilize relevant contextual information effectively. In contrast, GPT-3.5 Turbo without RAG shows lower and more variable performance, with notable gaps for several questions. These results emphasize the advantage of combining Llama 3.2 with RAG, as it delivers more accurate and contextually relevant responses compared to GPT-3.5 Turbo operating without retrieval support.

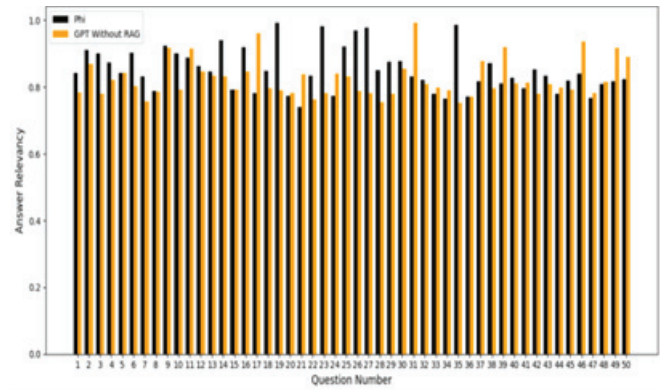


Fig. 4. Performance comparison of Microsoft Phi and GPT 3.5 turbo without RAG with Open AI Embedding

The results shown in Fig. 4 indicated that **Microsoft Phi 3.5 with RAG, which is given in black bars, outperforms GPT-3.5 Turbo without RAG in orange bars in terms of answer relevancy across the 50 questions.** Microsoft Phi 3.5 with RAG consistently achieves higher or comparable relevancy scores, with many scores reaching the maximum of 1.0, demonstrating its ability to deliver more accurate and contextually relevant answers by leveraging retrieved information. **GPT-3.5 Turbo without RAG, while consistent, exhibits relatively lower scores with noticeable gaps in performance for certain questions, emphasizing the effectiveness of RAG in enhancing answer quality with Microsoft Phi 3.5.**

VI. CONCLUSION

Health is a rapidly growing field where accessible assistance is essential, whether for students or general users. This project presents the development of a chatbot capable of interacting with users who provide data in PDF format, enabling efficient retrieval of any relevant information contained within. To enhance user convenience, the bot also incorporates voice interaction. The system has been evaluated using three different large language models paired with various embedding techniques. The analysis highlights the superior performance of GPT-3.5 Turbo with OpenAI embeddings, achieving top metrics in context precision, recall, faithfulness, and answer relevancy. This combination excels in both RAG and standalone evaluations, demonstrating exceptional reliability. Comparisons with LLaMA 3.2 and Microsoft Phi 3.5 show that RAG significantly enhances answer quality, as both models outperform GPT-3.5 Turbo without RAG. While LLaMA 3.2 ensures strong relevancy, its faithfulness is limited, leading to occasional inaccuracies. Microsoft Phi 3.5 balances precision and recall, making it a strong alternative. Among the various embedding techniques used, OpenAI embedding outperforms MiniLM and BGE in precision, recall and relevance though BGE offers competitive faithfulness with GPT-3.5 Turbo recall and relevance though BGE offers competitive faithfulness with GPT-3.5 Turbo. **From this study, the results show that a high-quality embedding and advanced LLMs improve answer reliability, making GPT-3.5 Turbo with OpenAI embeddings the best option for RAG.** Future advancements include fine-tuning embeddings with domain-specific data, hybrid approaches combining multiple LLMs dynamically and expanding RAG to multi-modal data for applications like voice assistants. **Enhancing real-time retrieval efficiency and handling complex queries will ensure scalability and industry**

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